

CIAN ROCHE

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EDUCATION

Massachusetts Institute of Technology <i>PhD candidate - Physics</i> Thesis: “ <i>The Dark Universe Through The Strong Lens: Cluster and Galaxy-Scale Probes of the Dark Sector</i> ” Advisors: Prof. Michael McDonald, Prof. Mark Vogelsberger	2021 – Present
Universität Tübingen <i>MSc Mathematical Physics</i> 1.1 GPA [†] Thesis: “ <i>Exact parallel waves in general relativity</i> ” Advisor: Prof. Carla Cederbaum	2019 – 2021
University College Cork <i>Bachelor of Science: Physics</i> , 1 st class honours	2015 – 2019

HONORS AND AWARDS

MIT Physics Henry W Kendall Teaching Fellowship	2026
DAAD Graduate Program Full Scholarship	2019
College Scholar - University College Cork	2019
Max Planck Institute for Plasma Physics Research Scholarship	2018
Naughton Research Fellowship (one of 8 in Ireland)	2017
Naughton Foundation Scholarship (one of 29 in Ireland)	2015
All-Ireland Scholarship (one of 125 in Ireland; declined)	2015

TEACHING

Massachusetts Institute of Technology	
Teaching Assistant, Experimental Physics II (8.14)	Spring 2026
Teaching Assistant, Experimental Physics I (8.13)	Fall 2024
Teaching Assistant, Experimental Physics I (8.13)	Spring 2024
Teaching Assistant, Experimental Physics I (8.13)	Fall 2023
Teaching Assistant, Experimental Physics I (8.13)	Fall 2022
Instructor - Gaia DR3 Hackathon	Jun 2022
University College Cork	
Teaching Assistant, Introductory Physics II	Spring 2018
Leader, Peer Assisted Learning Scheme	2016 – 2018

PUBLICATIONS

First Author or Significant Contribution

5. **Roche, C.**, Vogelsberger, M., McDonald, M., Dutra, I., Natarajan, P., Shen, X., Shreeram, X., Metcalf, R., Sharon, K., Birrer, S., Lee, W., Meneghetti, M. 2026; “*A Consistent Implementation of Cluster Strong Lensing in Cosmological Simulation Light Cones*”, [arXiv:2605.30433](#)
4. Tran, V., Shen, X., Vogelsberger, M., Gilman, D., O’Niell, S., **Roche, C.**, Zier, O., Gao, J. 2024; “*Novel Density Profile for Isothermal Cores of Dark Matter Halos*”, Phys. Rev. D 113, 063024, [arXiv:2411.11945](#)
3. **Roche, C.**, McDonald M., Vogelsberger, M., Shen, X., Springel, V., Hernquist, L., Pakmor, R., Bose, S., Kannan, R. 2024; “*Brightest Cluster Galaxy Offsets in Cold Dark Matter*”, OJA, astro-ph.GA 7, [arXiv:2402.00928](#)
2. **Roche, C.**, Necib, L., Lin, T., Ou, X., Nguyen, T. 2024; “*The Escape Velocity Profile of the Milky Way from Gaia DR3*”, ApJ, 972, 70, [arXiv:2402.00108](#)

[†]on a scale of 1-5, where 1 is the highest grade.

1. **Roche, C.**, Aazami, A.B. & Cederbaum, C. 2023; “*Exact parallel waves in general relativity*” Gen Relativ Gravit 55, 40, [doi:10.1007/s10714-023-03083-x](https://doi.org/10.1007/s10714-023-03083-x)

Nth Author

3. Rose J. C., et al. (incl. **Roche, C.**) 2025; “*The DREAMS Project: A New Suite of 1,024 Simulations to Contextualize the Milky Way and Assess Physics Uncertainties*”, [arXiv:2512.00148](https://arxiv.org/abs/2512.00148)
2. Nguyen, T., et al. (incl. **Roche, C.**) 2024; “*How DREAMS are made: Emulating Satellite Galaxy and Subhalo Populations with Diffusion Models and Point Clouds*”, ApJ, 997, 336, [arXiv:2409.02980](https://arxiv.org/abs/2409.02980)
1. Rose J. C., et al. (incl. **Roche, C.**) 2024; “*Introducing the DREAMS Project: DaRk mattEr and Astrophysics with Machine learning and Simulations*”, ApJ, 982, 68, [arXiv:2405.00766](https://arxiv.org/abs/2405.00766)

Non-Refereed

1. **Roche, C.** & Garnavich, P. 2020; “*Testing Progenitor Models Using the Late-Time Light Curve of Supernova 1992A*” Res. Notes AAS 4 207, [doi:10.3847/2515-5172/abcb05](https://doi.org/10.3847/2515-5172/abcb05)

PRESENTATIONS

Invited (Colloquia/Seminars)

Talk	“ <i>Light Cone Strong Lensing in Cosmological Hydrodynamical Simulations</i> ” MIT Kavli Institute for Astrophysics and Space Research	Mar 2026
Talk	“ <i>Understanding Dark Matter with Simulated Strong Lensing</i> ” Yale University	Sep 2025
Talk	“ <i>Dark Matter Self-Interaction via Galaxy Cluster Dynamics</i> ” Harvard University	Feb 2024
Talk	“ <i>Galaxies Lacking Dark Matter Produced in a Cosmological Simulation</i> ” MIT Kavli Institute for Astrophysics and Space Research	Apr 2022
Lecture	“ <i>Waves in General Relativity</i> ” Mathematical Relativity Class – University of Tübingen	Jul 2021
Talk	“ <i>Exact parallel waves in general relativity</i> ” Mathematical Physics Colloquium – University of Tübingen	Jun 2021
Talk	“ <i>How to Stop a Hurricane: Complexity in Physics</i> ” “Blow Your Mind Week” 2020 – University College Cork	Oct 2020
Talk	“ <i>Hermiticity of Young Operators</i> ” Young Tableaux Seminar Series – Universität Tübingen	Jun 2020

Contributed

Poster	“ <i>The Line of Sight as a Major Systematic of Cluster Strong Lensing</i> ” Gravitational Lensing at the Turning Point – Sesto	July 2026
Talk	“ <i>The Observation-Simulation Connection and the Line of Sight in Cluster Strong Lensing</i> ” Strong Gravitational Lensing Workshop – Montpellier	June 2026
Talk	“ <i>Consistent Strong Lensing Predictions from Cosmological Simulations</i> ” Scaling up Lensing Workshop – University of Liège	Oct 2025
Poster	“ <i>Unveiling Dark Matter: Testing CDM with the First End-to-End Cluster Strong Lensing Simulations</i> ” Tracing cosmic Evolution with Galaxy Clusters V – Sesto	Jul 2025
Talk	“ <i>Testing CDM with the First End-to-End Cluster Strong Lensing Simulations</i> ” European Astronomical Society Meeting 2025 – University College Cork	Jun 2025

Astrogazers

2024–Present

Massachusetts Institute of Technology

Volunteered for general public and youth outreach events with MIT astrogazers, including Massachusetts Space Week and Cambridge Science Festival.

MIT Physics Graduate Admissions

2022/2023 Cycle, 2023/2024 Cycle

Massachusetts Institute of Technology

Served for two admissions cycles on the admissions committee for the graduate program in physics as part of the SAMC (student admissions committee members) initiative.

AstroWiki

2023 – Present

Massachusetts Institute of Technology

Created and co-maintain a set of living notes covering most of graduate level astrophysics, specifically designed to aid the next generations of graduate students when studying for the MIT astrophysics oral qualification exam. ([link](#))

MIT Summer Research Program Admissions Committee

2022/2023 Cycle

Massachusetts Institute of Technology

Served on the admissions committee for the MIT Summer research program (MSRP), which aims in particular to provide high-quality research experiences to promising undergraduate students from underrepresented backgrounds.

MIT GAGA Initiative

Co-chair 2022 – 2023 | Organising Committee 2021 – 2024

Massachusetts Institute of Technology

The Graduates Advising Graduate Admissions (GAGA) program provides the graduate student perspective to the Chair of Graduate Admissions, offers information and mentorship to underrepresented groups in physics and advises on improvements to the admissions process.

MIT PhysGAAP program

2021 – 2023

Massachusetts Institute of Technology

The Physics Graduate Application Assistance Program (PhysGAAP) provides guidance to prospective applicants on navigating the application process, and liaises with the physics department leadership on issues of accessibility and equity in the graduate application. I have served as one of the primary organizers and have mentored 5 students through the program.

UCC Physics and Astronomy Society

2017 – 2019

University College Cork

Served as the Astronomy officer of the University College Cork Physics and Astronomy Society, organizing large astronomy outreach trips to dark sky regions in Western Ireland, and aurora borealis observation in Iceland.

Miscellaneous Volunteer Positions

Volunteer, AAS 241st Meeting Graduate School Fair 2023
 Desk Czar/building management liason, MIT Kavli Institute for Astrophysics and Space Research . 2022–2025
 Organizer, MIT Kavli Institute for Astrophysics and Space Research graduate student lunch 2022–2023
 Member, MIT Physics Graduate Student Council 2021 – 2023
 Organising Committee, MIT Graduate Student Union 2021 – 2023
 Student academic representative, University College Cork 2017 – 2019